



NHKG: A Validated Hybrid Pipeline for Hindi Event Knowledge Graph Construction

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[M.Tech Thesis Evaluation](#)

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Motivation and Problem

- Hindi text contains rich multi-faceted information about events, participants, and locations.
- Raw text remains difficult to query, formally validate, or reuse for logical reasoning.
- Hindi is a low-resource language for sophisticated, structured event extraction tasks.
- Problem: How can Hindi text be converted into a reliable, formally valid event-centric knowledge graph?

Objectives and Key Challenges

Core Objectives

- Extract events from sentences, enrich them using linguistic signals, construct a formal RDF knowledge graph, and evaluate both structural correctness and semantic extraction quality.

Key Challenges

- Trigger detection is ambiguous, generic predicates dominate, argument roles are hard to assign precisely, and the final graph must remain formally valid.

Research Evolution Across MTP

NHKG represents the integrated synthesis of multiple exploratory research tracks.

DBpedia Bootstrapping Translation & Linking Cross-Lingual OpenIE Neural Projection



WordNet Enrichment Taxonomy Benchmarks Grammar Prototypes Neuro-Symbolic Trials



Final NHKG System
Integrated Thesis

Lessons from Precursor Experiments

DBpedia Bootstrapping

- DBpedia bootstrapping showed that entity linking alone cannot capture event-centered knowledge.

Cross-Lingual OIE

- Cross-lingual OIE exposed transfer errors, tokenization mismatches, and weak Hindi predicate handling.

WordNet Taxonomy

- WordNet taxonomy work motivated lexical-semantic enrichment and controlled resource usage.

Grammar Prototype

- Grammar prototypes showed that free-form LLM extraction needs schema constraints.

Takeaway: These lessons shaped NHKG's final hybrid design: grounded candidates, constrained extraction, Hindi-specific refinement, and formal validation.

NHKG Architecture

Frame retrieval & candidate proposal.



Schema-constrained LLM extraction.



Hybrid candidate filtering & selection.



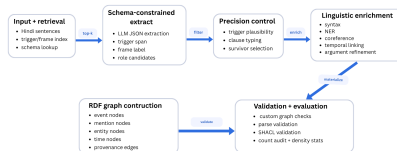
Linguistic enrichment.



**RDF graph construction,
reconciliation, and formal validation.**

NHKG Pipeline overview

From Hindi sentence input to validate event knowledge graph



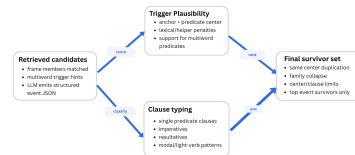
Key Design Choice: The LLM is used ONLY for schema-constrained event extraction. Deterministic code handles enrichment, graph construction, and validation.

Event Extraction Pipeline

- Frame candidates retrieved from UHVN/PropBank-style registry.
- Two-stage prompting: frame selection followed by frame-specific extraction.
- Extraction is strictly schema-constrained (JSON grammar).
- Post-extraction precision layer: trigger plausibility scoring, clause typing, same-center competition, and survivor selection.

Candidate extraction and survivor selection

How NHKG suppresses event explosion while keeping the best surviving candidates



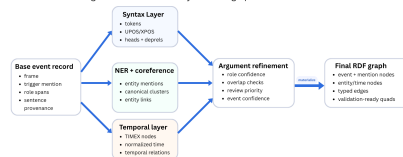
Key architecture bias: the system chooses the cleanest stable event anchor first, then keeps only the best surviving candidates

Enrichment and Construction

- Syntax layers: Dependency parsing & POS.
- Entity layers: NER & Coreference clustering.
- Temporal: Temporal enrichment, normalization, and event-time relation typing.
- Graph Materialization: Creates formal nodes (event, mention, entity, time) with strict role and provenance links.

Enrichment to graph construction

Post-extraction signals are fused into a layered RDF graph

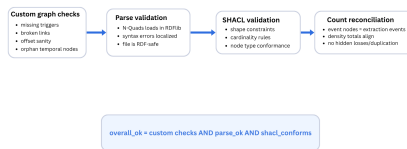


Graph Validation Layer

- Custom Structural Checks: Valid trigger/event links and correct text offsets.
- Parse Validation: Ensures the emitted RDF/N-Quads parses correctly.
- SHACL Validation: Graph conforms to formal schema constraints.
- Count Reconciliation: Extraction counts and graph counts remain exactly consistent across stages.

Graph validation pipeline

Structural correctness is checked before the graph is treated as a final research artifact



Evaluation Methodology

Structural Evaluation

Execution on a full 5,000-sentence run.
Focused on exact parse validation, strict SHACL conformance, count reconciliation, and graph density behavior.

Silver Semantic Evaluation

Adjudicated sample analysis of emitted events and audit of missed-event cases. Measured event reality, trigger correctness, frame plausibility, and exact argument usability.

Final Structural Results

Metric	Value
Input sentences	5000
Event-bearing sentences	2568 (51.36%)
Extracted events	2705
Events per input sentence	0.5410

Check	Result
overall_ok	True
parse_ok	True
shacl_conforms	True
Count mismatch	0

Takeaway: NHKG scaled successfully and the final graph remained formally valid and auditable.

Silver Semantic Evaluation

From 300 adjudicated emitted-event items:

Measure	Rate
Event real rate	87.67%
Trigger correct rate	50.00%
Frame plausible rate	41.67%
Arguments usable rate	27.67%
Duplicate/alternate rate	26.67%
Overall usable event rate	23.67%

Many emitted items are genuinely eventive, but precise trigger, frame, and argument quality remains limited.

Missed-Event Audit & Errors

Error concentration: generic frame dominance, weak trigger specificity, poor argument usability, and conservative behavior on subtle eventive constructions.

Audit (N=50 no-event items):

- 66% should have had an event (30% true no-event, 4% uncertain).

Most common missed-event families:

- change_of_state: 13
- modal_lexical: 12
- light_verb_compound: 8
- imperative: 6
- state: 5
- action: 4

Prompt-Side Diagnostics

These experiments were useful diagnostically, but not stable enough to replace the frozen NHKG pipeline.

- Experimental Goal: Test if prompt changes alone improve semantic extraction (run outside core pipeline).
- Context_rich variant: Added syntax/WordNet context. Changed local behavior but did not consistently improve frame quality.
- Predicate_contrastive variants: Explored stricter joint predicate-frame selection. v1 caused recall collapse; v2 recovered 2 of 3 baseline-positive diagnostic cases, but remained exploratory.
- Discovery: Diagnostics revealed a builder-side predicate-card coverage failure.

Key Contributions and Limitations

Contributions

Delivered an end-to-end validated Hindi event KG pipeline integrating schema-constrained extraction, deterministic enrichment, provenance, and formal validation.

Limitations

Low trigger/frame precision and weak argument usability. The pipeline remains conservative, leading to significant missed-event recall gaps on subtle and complex eventive constructions.

Conclusion

- Final output: a complete, validated Hindi event-centric knowledge graph produced from raw Hindi text.
- NHKG establishes a solid infrastructure for Hindi event-centric knowledge graphs, with semantic improvement as the next natural research step.
- NHKG successfully builds a validated, reproducible, and scalable Hindi event knowledge graph pipeline.
- The strongest contribution is structural correctness and graph construction.
- The main limitation is semantic event quality.

Thank You / Questions

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